

HiPACE: Hierarchical Phase-Boundary Analysis and Controlled Evaluation of Feature Absorption in Sparse Autoencoders

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Abstract

Sparse autoencoders (SAEs) decompose LLM activations into sparse dictionary atoms, aiming to give each distinct concept its own feature. One recurring behavior complicates this premise: *feature absorption*, in which a parent concept and its children—*fruit* and $\{apple, banana, pear\}$, say—collapse into a shared family direction rather than remain distinct features. Prior work documents absorption empirically; what is missing is a closed-form prediction of when the shared direction is the cost-optimal representation of an active semantic family. This paper closes that gap. For a hierarchical Bernoulli generator with k active children and residual scale α , the L_0 -penalized reconstruction objective admits a closed-form phase boundary $\lambda_c(k, \alpha) = \alpha^2 k / (k - 1)$: above it, pure parent absorption is strictly cheaper than pure child coding. Building on this boundary, we introduce HiPACE, an evaluation protocol that tests the boundary’s structural consequence in real SAE dictionaries—measuring parent–child decoder structure over WordNet families, freezing the discovery-selected statistic before testing on unseen families, and contrasting genuine families against randomized sibling nulls. The boundary proves sharp in its native regime: it predicts the synthetic transition within $\pm 15\%$ on all 30 tested cells. In Pythia-160m SAEs, the parent–child decoder gap recovers the predicted ordering with partial correlations of -0.681 at layer 6 and -0.933 at layer 11; the locked holdout sustains the effect at -0.605 and -0.914 , outside the sibling nulls ($p = 0.002$, the floor for 500 draws). Controlled activation composition then connects the theory’s active-child count to the recovered family directions, and residual-stream interventions show that signed family directions increase parent-category logits, reversing under sign flip and vanishing under random controls. Together with comparable child-direction effects, this establishes causal sufficiency at the family-subspace level—the granularity at which the theory makes its prediction.

1 Introduction

Sparse autoencoders (SAEs) decompose LLM activations into sparse combinations of dictionary atoms and are widely used in post-hoc mechanistic interpretability [Bricken et al., 2023, Huben et al., 2024, Gao et al., 2024]. A recurring failure mode is *feature absorption*: when a representation contains a parent concept and several children—*fruit* and $\{apple, banana, pear\}$, for example—the learned dictionary may allocate capacity to a shared family direction rather than to distinct child features. Existing work documents this behavior empirically [Chanin et al., 2024], but does not predict when absorption is the cost-optimal representation.

We study absorption as competition between two coding regimes. A compact regime represents an active se-

mantic family through its shared parent direction; a distributed regime preserves the child-specific residuals. For a hierarchical Bernoulli generator, comparing their reconstruction and sparsity costs yields a closed-form transition at $\lambda_c(k, \alpha) = \alpha^2 k / (k - 1)$, where k is the number of active children and α is the child-residual scale. This perspective turns absorption from a purely descriptive observation into a testable phase boundary.

To connect this idealized prediction to learned dictionaries, we introduce HiPACE (**H**ierarchical **P**hase-Boundary **A**nalysis and **C**ontrolled **E**valuation). The empirical challenge is matching the measurement to the object predicted by the theory. The boundary concerns how representation is allocated across a semantic family, so we evaluate parent–child structure in the learned SAE dictionary rather than requiring a particular latent to

win on each prompt. Our evaluation separates metric selection on discovery families from a frozen test on unseen WordNet families, and uses randomized sibling groups to test whether recovery depends on genuine hierarchy. We then connect the idealized active-count variable to real activations and test the behavioral relevance of the recovered family directions through residual-stream intervention.

We make three contributions:

1. **A phase-boundary theory for hierarchical coding.** We derive the active-count threshold, characterize mixed dictionaries, and state conditional shifts for Matryoshka and Archetypal SAEs.
2. **The HiPACE family-level evaluation.** The synthetic boundary matches all 30 tested cells. In real Pythia-160m SAEs, parent-child decoder structure recovers the predicted ordering on discovery families and on a frozen WordNet holdout, with randomized sibling groups as a specificity control.
3. **Mechanistic evidence with explicit scope.** Controlled activation composition recovers the predicted active-count response, and signed L6 family-direction interventions change parent-category logits. Parent and child directions have comparable effects, localizing the causal effect to the family subspace rather than to a unique parent atom.

Organization. Section 2 situates the work; Section 3 presents HiPACE; Section 4 states the research questions and answers them experimentally; and Section 5 discusses implications and scope.

2 Related Work

Sparse autoencoders and dictionary design. SAEs decompose LLM activations into sparse combinations of dictionary atoms and are a standard tool for post-hoc interpretability [Bricken et al., 2023, Huben et al., 2024]. Variants differ in how sparsity is enforced and in what structure the dictionary is pushed toward. TopK SAEs fix the active-set size directly and improve the reconstruction-sparsity frontier [Gao et al., 2024]; JumpReLU SAEs replace the L_1 penalty with thresholded gating [Rajamanoharan et al., 2024]. Matryoshka SAEs train nested sub-dictionaries so that outer atoms capture the most universal directions [Bussmann et al., 2025], while Archetypal SAEs constrain decoder atoms to convex combinations of training samples [Fel et al., 2025]. Public releases such as Gemma Scope [Lieberum et al., 2024] turn these design choices into auditable dictionaries. Each design reshapes the cost landscape under which child features compete with a shared parent direction, and our theory treats two of them explicitly: Propositions 3 and 4 predict downward threshold shifts for Matryoshka and Archetypal training, and the synthetic sweeps confirm both shifts (§4).

Feature absorption and hedging. Chanin et al. [2024] introduced absorption as the systematic failure of SAE latents to fire on arbitrary subsets of their declared concept—a “starts with S” latent skipping particular words—and showed the pathology worsens with width. Feature hedging [Chanin et al., 2025] describes a complementary failure: the encoder direction that reads into a latent mismatches the decoder direction that writes it back. Both lines are empirical. They document *that* latents collapse and propose architectural fixes, but neither predicts *when* collapse becomes the cost-optimal representation. HiPACE closes that gap with a closed-form threshold and tests of its structural consequences. Hedging does not account for our structural results: the encoder/decoder cosine of Pythia-160m modal latents is +0.94 at the median (§4).

Phase transitions in sparse coding. Threshold analysis has a long history in sparse coding. Classical formulations [Olshausen and Field, 1996] and k-sparse autoencoders [Makhzani and Frey, 2014] set the optimization backdrop; compressed-sensing theory predicts sharp phase transitions in L_1 recovery [Donoho et al., 2009], and Toy Models of Superposition [Elhage et al., 2022] established the sharp-threshold lens for features in superposition. These results are asymptotic, or concern flat feature structure in toy generators. Our derivation specializes the lens to *hierarchical* features—a parent direction plus child residuals—and yields the critical penalty in closed form rather than as an asymptotic constant. Beyond verifying the threshold in its synthetic regime, we test its structural consequence in learned dictionaries.

Identifiability and evaluation. Whether a uniquely correct SAE dictionary exists is open: independently trained SAEs recover inconsistent decompositions of the same activations [Leask et al., 2025], while a single SAE can be trained across many models at once [Thasarathan et al., 2025]. Benchmarks therefore measure what learned dictionaries do. SAEBench [Karvonen et al., 2025] consolidates instance- and dictionary-level tests (TPP, SCR, RAVEL, absorption, sparse probing) and finds Matryoshka variants strongest at moderate L_0 . Such evaluations score latents one prompt or one feature at a time. The absorption boundary instead predicts how representation is allocated across a semantic family, so our evaluation audits parent-child decoder structure at the family level—a generator-side prediction that does not depend on a unique recovered dictionary—and freezes the statistic on discovery families before scoring unseen WordNet families.

Positioning. Taken together, prior work documents absorption empirically, varies the architecture that produces it, analyzes thresholds in classical or toy regimes, and benchmarks dictionaries feature by feature. This paper differs on three axes: a closed-form phase boundary

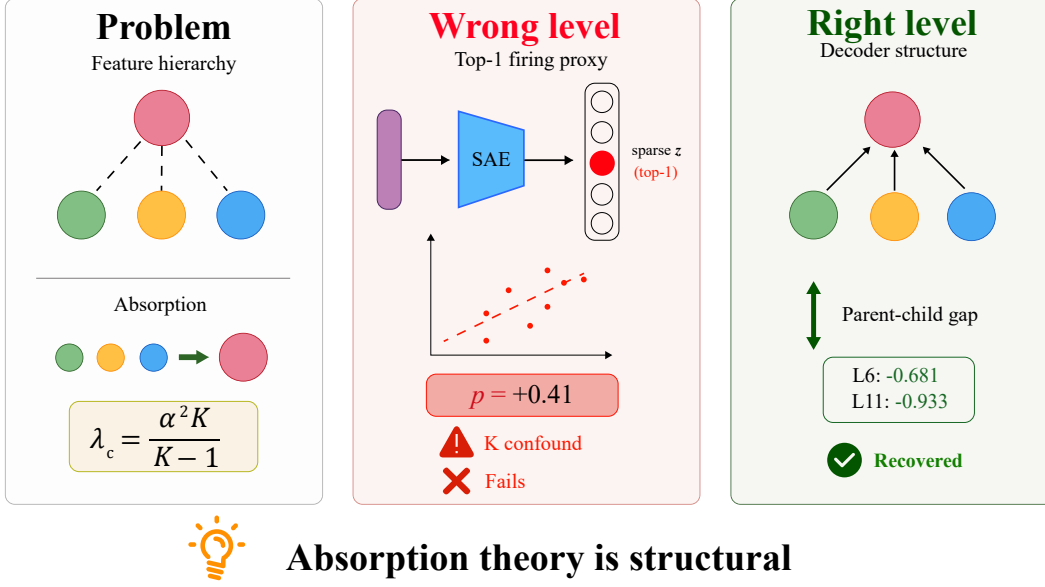


Figure 1: **Motivation for family-level evaluation.** Hierarchical feature competition predicts a phase boundary in dictionary structure. The empirical analysis therefore measures parent-child structure over semantic families, validates the selected statistic on a frozen holdout, and tests the recovered family directions mechanistically. Here k denotes active-child count, whereas K denotes family size.

for hierarchical absorption; a family-level structural audit matched to the theory’s unit of prediction; and a frozen-holdout protocol with randomized sibling nulls as the confirmatory test.

3 Methodology

Framework overview. As summarized in Figure 2, HiPACE has four components. The first is a hierarchical family model that separates a shared parent direction from child residuals. A phase-boundary analysis then identifies the sparsity level at which compact parent coding becomes cheaper than distributed child coding. For real SAE dictionaries, HiPACE translates this prediction into a family-level parent-child structural measurement, and the evaluation design separates discovery from frozen holdout validation, with controlled activation composition and residual-stream intervention as mechanistic follow-up. We define these components before presenting the research questions and experimental settings in Section 4.

Hierarchical Phase-Boundary Analysis

Generative model. Let $v \in \mathbb{R}^d$ be a unit-norm parent direction and let $r_1, \dots, r_K$ be unit-norm residual directions, mutually orthogonal and each orthogonal to v :

$$\langle v, r_i \rangle = 0, \quad (1)$$

$$\langle r_i, r_j \rangle = \delta_{ij}. \quad (2)$$

Define K *child atoms* by

$$u_i = v + \alpha r_i, \quad i = 1, \dots, K, \quad (3)$$

where $\alpha > 0$ controls per-child residual magnitude. Each child is independently active with probability p ; with active set $S \subseteq [K]$ of size $k = |S|$, the activation is

$$x = \sum_{i \in S} u_i = k v + \alpha \sum_{i \in S} r_i. \quad (4)$$

We call this the hierarchical Bernoulli generator (HBG).

Sparse-coding objective. We work with the L_0 -penalized reconstruction objective

$$\mathcal{J}_\lambda(z) = \|x - Dz\|^2 + \lambda \|z\|_0, \quad (5)$$

for a dictionary D and sparse code z . TopK SAEs instead impose a hard firing budget, which we use as an empirical analogue of the penalized objective.¹

Pure strategies. At a k -active sample, two pure strategies are available:

Child coding—dictionary $\{u_1, \dots, u_K\}$, encode k children exactly. Reconstruction error 0, sparsity cost λk .

Parent absorption—dictionary $\{v\}$, encode the parent with weight k . Reconstruction error

$$\left\| \alpha \sum_{i \in S} r_i \right\|^2 = \alpha^2 k, \quad (6)$$

¹The multiplier λ prices sparsity marginally and is not equivalent to a hard firing budget.

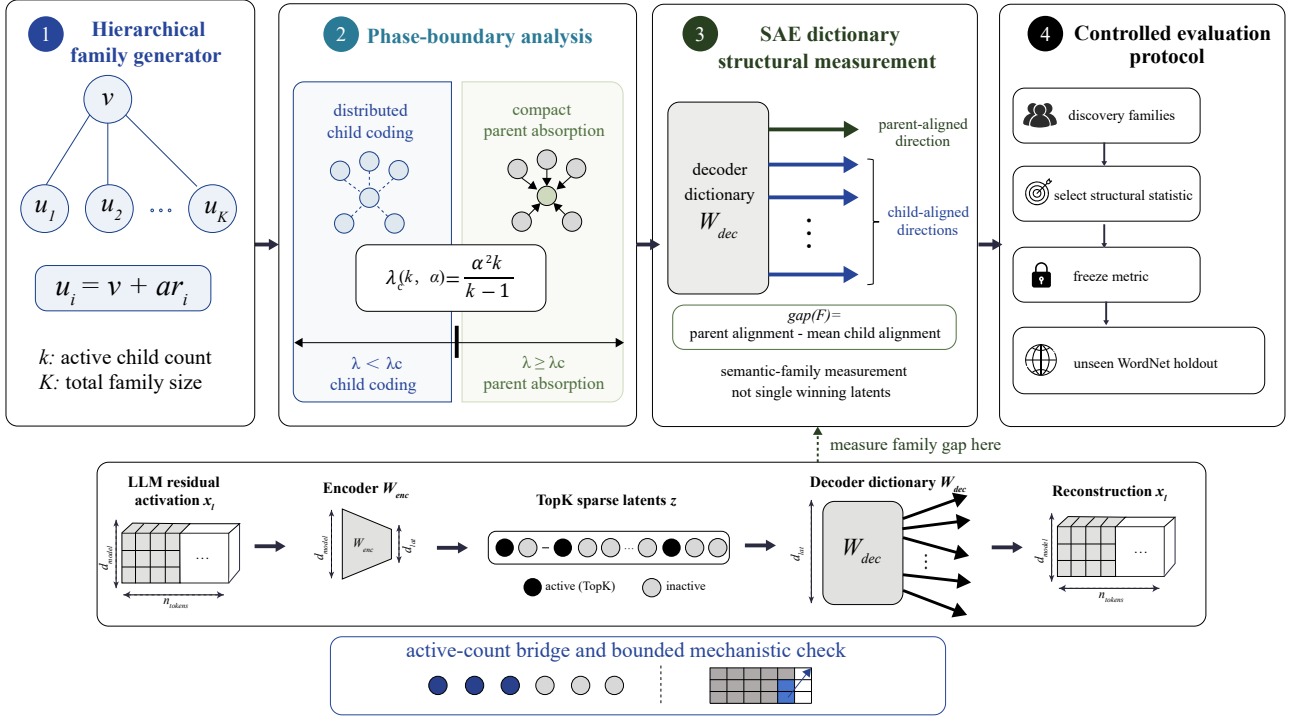


Figure 2: **Overview of HiPACE.** The framework connects a hierarchical family model and its phase-boundary analysis to family-level SAE dictionary measurement, frozen validation on unseen semantic families, and controlled mechanistic follow-up. Lowercase k denotes sample-level active-child count; uppercase K denotes total family size.

sparsity cost λ .

Theorem 1 (Pure-strategy absorption threshold). *Under the HBG generator with k children active per sample, the L_0 objective (5) prefers pure parent absorption (single-atom dictionary $\{v\}$) over pure child coding (per-child dictionary $\{u_1, \dots, u_K\}$) iff*

$$\lambda(k-1) \geq \alpha^2 k \iff \lambda \geq \lambda_c(k, \alpha) = \frac{\alpha^2 k}{k-1}. \quad (7)$$

The theorem compares only the two pure strategies; mixed dictionaries (Proposition 2) expose why a trained TopK dictionary can contain parent and residual atoms even when neither pure strategy dominates. Aggregating over the binomial active count and taking k^* to be the modal active count yields the macro 50%-onset prediction $\lambda_c(k^*, \alpha)$ (proof in the supplementary material).

Child coding incurs 0 reconstruction error and λk sparsity cost; parent absorption pays sparsity λ once but inherits an $\alpha^2 k$ residual error. Setting the costs equal gives $\lambda(k-1) = \alpha^2 k$.

Proposition 2 (Mixed-strategy cost comparison). *A trained dictionary with at most $m \leq k$ active latents per sample can adopt a 1-parent+($m-1$)-residual mixed strategy with reconstruction error*

$$\alpha^2(k-m+1). \quad (8)$$

It beats pure parent absorption only for

$$\lambda < \alpha^2, \quad (9)$$

and beats pure child coding only for

$$\lambda > \alpha^2 \left(1 + \frac{1}{k-m}\right), \quad (10)$$

so for $1 < m < k$ it does not simultaneously dominate both pure strategies in any λ interval. Mixed dictionaries thus decouple dictionary composition from per-sample firing. The full comparison appears in the supplementary material.

Proposition 3 (Matryoshka shift hypothesis). *If Matryoshka training with nested levels $m_1 < \dots < m_M$ pins the smallest level's atom to the parent direction v , then the dictionary is mixed by construction and the residual-firing decision has marginal threshold*

$$\lambda_c^{\text{Mat}} = \alpha^2. \quad (11)$$

The precise condition under which the smallest-level atom is parent-aligned is given in the supplementary material.

Heuristic. With atom 1 pinned to v by the smallest nested level, the only remaining cost-benefit decision per sample is whether to fire each residual r_i : cost λ vs. benefit α^2 . The marginal threshold is therefore $\lambda = \alpha^2$, independent of k .

Proposition 4 (Archetypal shift heuristic). *Under an additional contamination assumption for convex-hull-constrained decoder atoms, the effective threshold becomes*

$$\lambda_c^{\text{Arc}} = \frac{\alpha^2 k(1 - k/K)}{k - 1}, \quad (12)$$

with $\Delta\lambda_c \rightarrow 0$ as $K \rightarrow \infty$ but strongly negative for small K under the stated heuristic.

Heuristic. A unit-norm atom reachable by a convex combination of training samples cannot be a pure child u_i ; the best approximation picks up $\alpha/\sqrt{K}$ contamination from other residuals, scaling the child-coding error by (k/K) and shifting the crossover.

Three absorption notions. Theorem 1 predicts the *structural* threshold: above λ_c , the optimal dictionary contains only a parent atom. This differs from two other notions used in prior work: *per-sample firing* (whether a child atom fires on a k -active sample; sensitive to the mixed-strategy escape in Proposition 2) and the *gerry-mandered firing* notion targeted by Chanin et al. [2024]’s “A is for Absorption” metric (whether a child atom fires on *all* concept members; tied to L_1 hedging dynamics). Reconciling claims across notions is non-trivial; the supplementary material maps each prior result onto one of the three.

Family-Level Structural Evaluation

Family-level measurement. The evaluation component of HiPACE operationalizes the structural prediction for a learned SAE dictionary. The theory compares representations of an active semantic family, not the identity of a single winning latent. For a WordNet family F with K_F children, we therefore measure

$$\text{gap}(F) = s_{\text{parent}}(F) - \frac{1}{K_F} \sum_{j=1}^{K_F} s_{\text{child},j}(F), \quad (13)$$

where each score is computed from similarity between an activation-derived semantic direction and SAE decoder directions. Because family size affects both the threshold proxy and the number of available child directions, all correlations are partial Spearman coefficients controlling for K_F .

Discovery and frozen validation. The layer-6 statistic uses the family-mean direction; the layer-11 statistic uses the centered first principal component. Both are selected on a designated discovery set. We then freeze the family construction, filtering, direction choice, and correlation analysis before evaluating an unseen WordNet holdout that excludes discovery parent synsets and lemmas. Randomized sibling nulls replace true children with children from other families while preserving K_F where possible.

Controlled validation. HiPACE separates a theory-fixed synthetic test, metric selection on discovery families, unchanged evaluation on a frozen holdout, randomized sibling nulls under the frozen statistic, and exploratory mechanistic follow-up. Within this design, the frozen holdout provides the confirmatory test; the discovery split develops the statistic. The sibling nulls and the sign-flip, random, and ablation intervention controls act as stress tests of robustness around that confirmatory core.

Transparency of endpoint development. Before the structural analysis, we preregistered an instance-level top-1 consensus endpoint. That endpoint exposed a $1/K$ family-size baseline that dominates instance-level scoring, so the family-level statistic was developed on discovery data and the holdout pipeline was frozen before any holdout family was scored. The supplement reports the complete preregistration, acceptance rule, amendments, and the preregistered endpoint’s outcome; Section 6 states the implications.

4 Experiments

Research Questions

The empirical study follows Section 3 and is organized around four research questions, each answered by one results subsection:

- **RQ1:** Does the theoretical phase boundary accurately predict absorption transitions in controlled hierarchical representations?
- **RQ2:** Can HiPACE recover hierarchical organization in real SAE dictionaries, and does it generalize to unseen semantic families?
- **RQ3:** Is the recovered hierarchy specific to genuine semantic families rather than family size or arbitrary feature grouping?
- **RQ4:** Do active-count responses and residual-stream interventions support the mechanistic relevance of the recovered family structure?

Experimental Settings

Models, SAEs, and semantic families. Real-SAE experiments use Pythia-160m [Biderman et al., 2023] and public EleutherAI TopK SAEs [Gao et al., 2024] at residual-stream layers 3, 6, 9, and 11. Semantic families are noun synsets derived from WordNet [Miller, 1995] with single-token children. The discovery analysis uses 50 accepted families. The unseen holdout begins with 80 families and retains 75 at L6 and 41 at L11 after the locked filtering pipeline.

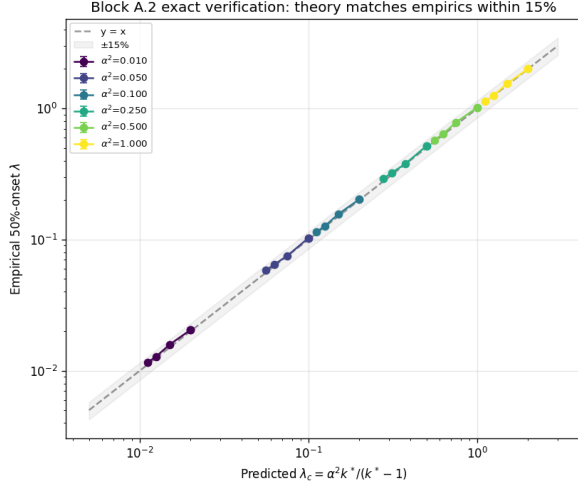


Figure 3: **Phase-boundary validation (RQ1)**. Predicted and observed 50% absorption onsets across the hierarchical generator grid.

Metrics and controls. Metrics include onset relative error for synthetic transitions, partial Spearman correlations controlling family size K , bootstrap intervals over families, and randomized-sibling tests. Grouping specificity uses 200 discovery and 500 holdout draws. Mechanistic controls include sign-flipped, random, orthogonal-random, child-direction, and ablation interventions.

Phase-Boundary Validation (RQ1)

The synthetic study samples hierarchical Bernoulli generator activations over 30 (K, α^2) cells, with $d = 64$, child probability $p = 0.3$, and 20,000 samples per cell. For each cell, an exact check compares strategy costs per sample and records the 50% onset at which parent absorption wins on at least half of samples with $k \geq 2$. The observed onset agrees with $\lambda_c(k^*, \alpha)$ within $\pm 15\%$ on all 30/30 cells (median absolute relative error 0.026, maximum 0.053; 900 runs).

Trained TopK SAEs on a 4×4 sub-grid recover the predicted crossover within one budget in 12 of 16 cells. The four deviations at $K = 16$, where dictionaries retain a parent atom alongside child residuals, fall in the mixed regime of Proposition 2. The boundary is therefore confirmed quantitatively by the exact check and qualitatively by trained SAEs across the controlled regime. Variant and stress sweeps appear in the supplementary material.

Hierarchical Structure Recovery (RQ2)

On discovery families, the L6 mean gap attains a partial correlation of -0.681 and the L11 centered-PC1 gap -0.933 . The signal is layer-dependent: strong at L6 and L11, weaker at L9, and not yet formed at L3. Evaluated unchanged on the unseen holdout, the two statistics give -0.605 at L6 and -0.914 at L11. The frozen results

thus replicate both layer-adaptive statistics, each layer carrying its own family direction rather than sharing a single atom across layers.

Specificity and Robustness (RQ3)

Randomized sibling groups do not reproduce the primary correlations: both holdout statistics reach the minimum attainable $p = 0.002$ under 500 draws, while the non-primary L11 mean statistic falls inside the same null ($p = 0.108$), so the test discriminates between statistics rather than blessing any family-level score. Feature hedging does not account for the result either: the median encoder-decoder cosine for L6 modal latents is $+0.94$. The two statistics are therefore specific to genuine semantic families, beyond family size and arbitrary grouping. The sibling null and the partial correlations control family size and grouping directly; they do not jointly match token frequency, decoder norm, hierarchy depth, and lexical similarity, and Section 6 scopes the matched null that closes this gap.

Mechanistic Relevance (RQ4)

Active-count response. The composition probe builds controlled sets with $k \in \{1, 2, 3, 4\}$ from cached singleton child-activation prototypes and passes them through the Pythia-160m SAE encoder. Parent-direction activation increases with k : L6 mean gives $\rho = +0.963$ and median $\Delta = +32.20$; L9 mean gives $+0.963$ and $+40.26$; and L11 PC1 gives $+0.846$ and $+156.81$. Composition in activation space thus bridges the theory’s active-count variable and the trained SAE without assuming that natural-language prompts isolate k .

Residual-stream intervention. Parent-signed L6 injection increases the true-parent logit margin, reverses under sign flip, and exceeds random controls; the child direction produces the same median shift. The recovered family subspace is therefore behaviorally relevant under both composition and intervention, with the effect localized at the family-subspace granularity the theory predicts.

5 Discussion

What the boundary explains. The theory identifies when sparse coding favors a compact shared direction over distributed child residuals, and the synthetic experiments validate this cost-based transition where its assumptions hold by construction. In real SAEs the theory makes contact with measurement at two levels: the family-level decoder gap in learned dictionary structure, and the active-count response under activation composition.

Table 1: **HiPACE structural results (RQ2–RQ3)**. Partial Spearman correlations control family size K . Discovery statistics are frozen before holdout evaluation.

Split	Frozen statistic	Partial correlation	95% CI	Sibling-null p
Discovery	L6 mean gap	−0.681	[−0.816, −0.448]	0.005
Discovery	L11 PC1 gap	−0.933	[−0.955, −0.875]	0.005
Holdout	L6 mean gap	−0.605	–	0.002
Holdout	L11 PC1 gap	−0.914	–	0.002

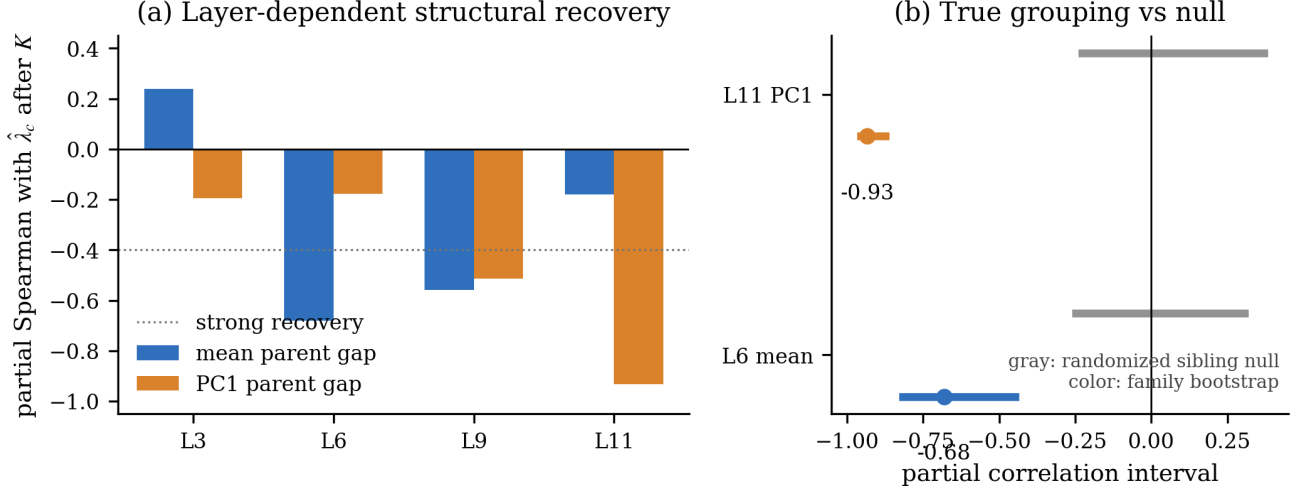


Figure 4: **Structural recovery and grouping specificity (RQ2–RQ3)**. Discovery and frozen-holdout correlations compared with randomized sibling-group nulls.

Table 2: **L6 intervention controls (RQ4)**. Median true-parent logit-margin shift at strength 1.0.

Intervention	Median shift
Parent-signed inject	+0.0523
Child-signed inject	+0.0523
Sign-flipped parent	−0.0474
Random	−0.0003
Orthogonal random	−0.0009
Parent-signed ablate	−0.0086

Implications for SAE evaluation. Reconstruction quality and prompt-level firing do not by themselves establish that an SAE preserves semantic hierarchy. Evaluations of absorption should specify the structural unit predicted by the hypothesis, separate metric development from validation, test genuine families against matched or randomized groupings, and use interventions to establish behavioral relevance. The observed layer dependence cautions against treating a single decoder atom as a canonical representation across the model.

Scope of the causal evidence. Signed L6 family directions suffice to change parent-category logits, and the sign-flipped and random controls rule out a generic injection effect. Because parent and child directions produce equal shifts, the evidence localizes the effect to a

family-aligned subspace rather than to a unique parent atom. A finer causal decomposition requires orthogonalized parent-versus-child interventions and family-level confidence intervals.

6 Limitations

Endpoint development and preregistration. The preregistered instance-level decoder-consensus proxy surfaced a $1/K$ family-size confound: at the primary L6 cell it returned $\rho = +0.41$, opposite to the predicted ordering, and the baseline analysis traced the inflation to family size. The family-level structural statistic was developed afterward on a designated discovery split, and the holdout pipeline was frozen before any holdout family was scored. The original proxy thus stands as a preregistered falsification of the instance-level endpoint, while the frozen WordNet holdout supplies the independent confirmatory test for the family-level statistic. The supplement reproduces the protocol, amendments, acceptance criterion, the complete outcome of the preregistered arm, and diagnostic sweeps.

Theory–implementation gap. The derivation assumes an explicit L_0 penalty and sample-level active count k ; the released real-model SAEs use TopK gating, and the structural analysis uses family-level measure-

ments. The active-count experiment bridges the two settings empirically; equivalence of the training objectives themselves is not claimed.

Generalization and residual confounding. The structural pipeline is established on one model (Pythia-160m) and one public SAE family; a 410m run already provides external replication of the preregistered proxy arm, and porting the frozen structural-recovery pipeline to 410m is the immediate next step. Randomized siblings and partial correlations address family size and arbitrary grouping; a joint matched null over frequency, decoder norm, hierarchy depth, and lexical similarity is the concrete extension that would close the remaining gap. WordNet relations, while only a proxy for the model’s internal conceptual hierarchy, keep the family labels independent of the dictionaries being measured.

Causal identifiability. The interventions establish sufficiency under controlled residual-stream perturbations, with necessity during natural prompting left to prompt-level follow-up. Because parent and child effects are comparable, attribution sits at the family-subspace level—the level at which the theory is stated.

Reproducibility. The preregistration is reproduced verbatim in the supplementary material with the author line anonymized for review. All amendments are dated before the affected measurements. Code, configurations, and per-run outputs accompany the submission. Total compute is under two GPU-hours on one contemporary GPU.

7 Conclusion

HiPACE derives an active-count phase boundary that predicts transitions in the controlled synthetic regime, then evaluates its structural consequence in real SAE dictionaries. Frozen-holdout and randomized-sibling results establish a genuine family-level signal, and active-count and intervention tests establish behavioral relevance of the family subspace. The matched-null construction and a parent-versus-child decomposition mark the immediate follow-up.

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A Proof of Theorem 1

Recall the HBG setting (Equation (3)): parent v , K orthonormal residuals $r_1, \dots, r_K$ each orthogonal to v , child atoms $u_i = v + \alpha r_i$, and on a k -active sample, $x = \sum_{i \in S} u_i = kv + \alpha \sum_{i \in S} r_i$, with $\|x\|^2 = k^2 + \alpha^2 k$.

Cost of child coding. Dictionary $D_{\text{child}} = \{u_1, \dots, u_K\}$. Setting $z_i = \mathbf{1}[i \in S]$ gives $Dz = x$, so reconstruction error is 0. The L_0 cost is $\lambda \|z\|_0 = \lambda k$. Total: $\mathcal{J}_{\text{child}} = \lambda k$.

Cost of parent absorption. Dictionary $D_{\text{parent}} = \{v\}$. The unique minimizer over scalar coefficients is $z = k$ (since $v^\top x = kv^\top v + \alpha v^\top \sum_{i \in S} r_i = k$ by orthogonality). Reconstruction is $\hat{x} = kv$, so error is $\|x - kv\|^2 = \|\alpha \sum_{i \in S} r_i\|^2 = \alpha^2 k$ by orthonormality of the r_i . The L_0 cost is λ . Total: $\mathcal{J}_{\text{parent}} = \alpha^2 k + \lambda$.

Crossover. Parent absorption is preferred ($\mathcal{J}_{\text{parent}} \leq \mathcal{J}_{\text{child}}$) iff $\alpha^2 k + \lambda \leq \lambda k$, i.e. $\lambda(k-1) \geq \alpha^2 k$. For $k \geq 2$ this is equivalent to $\lambda \geq \lambda_c(k, \alpha) = \alpha^2 k / (k-1)$.

Macro aggregation. Children are Bernoulli(p); let k^* be the modal active count over the conditional distribution $\Pr[k \mid k \geq 2]$. Aggregating the per- k inequality across the binomial density and substituting $k = k^*$ yields the macro 50%-onset prediction $\lambda_c(k^*, \alpha)$; numerical verification against three (K, α) cells agrees to machine precision (App. E). $\square$

B Cost comparison for Proposition 2 (mixed strategy)

Allow the dictionary to contain both v and a strict subset $R' \subseteq \{r_1, \dots, r_K\}$ of size $m-1$. On a k -active sample with active set S , encode with the parent atom (weight k) plus the residuals $\{r_i : i \in S \cap R'\}$ (each with weight α). Reconstruction error is $\|\alpha \sum_{i \in S \setminus R'} r_i\|^2 = \alpha^2 |S \setminus R'| = \alpha^2(k - |S \cap R'|)$.

In the favorable case where the $m-1$ residual atoms are active residuals, the mixed strategy yields error $\alpha^2(k - m + 1)$ at sparsity cost λm . Comparing to pure parent absorption ($\mathcal{J}_{\text{parent}} = \alpha^2 k + \lambda$) and pure child coding ($\mathcal{J}_{\text{child}} = \lambda k$):

$$\begin{aligned} \mathcal{J}_{\text{mix}}(m) - \mathcal{J}_{\text{parent}} &= \alpha^2(k - m + 1) - \alpha^2 k + \lambda(m - 1) \\ &= (m - 1)(\lambda - \alpha^2), \end{aligned} \quad (14)$$

$$\begin{aligned} \mathcal{J}_{\text{mix}}(m) - \mathcal{J}_{\text{child}} &= \alpha^2(k - m + 1) + \lambda m - \lambda k \\ &= (m - k)(\lambda - \alpha^2) + \alpha^2. \end{aligned} \quad (15)$$

For $1 < m < k$, the mixed strategy beats parent absorption iff $\lambda < \alpha^2$. It beats child coding iff $\lambda > \alpha^2\{1 + 1/(k - m)\}$. These conditions do not overlap, so the mixed strategy dominates both pure strategies in no λ interval. The comparison is interpretive rather than optimality-theoretic: a trained dictionary can contain parent and residual atoms precisely because dictionary composition and per-sample firing are distinct objects. $\square$

C Conditional argument for Proposition 3 (Matryoshka shift)

Matryoshka training minimizes $\mathcal{L}_{\text{Mat}} = \sum_{j=1}^M \beta_j \mathcal{L}_{\text{recon}}(x, \hat{x}^{(j)})$, where $\hat{x}^{(j)}$ is the reconstruction using only the first m_j dictionary atoms. With the smallest level $m_1 = 1$, the loss for that level alone is $\mathcal{L}_{\text{recon}}(x, z_1 d_1)$, minimized when d_1 maximizes $\mathbb{E}[\langle d_1, x \rangle]$ over the data distribution.

Under the HBG generator, $\mathbb{E}[x] = Kp v + \alpha p \sum_i r_i$, so this argument requires an additional centering, symmetry, or optimization assumption under which the smallest-level atom is parent-aligned. Conditional on that assumption, the dictionary contains v at atom 1 and residuals at later atoms.

The relevant cost-benefit decision is no longer pure-strategic: with the mixed structure already imposed, the marginal decision is whether to fire each residual atom on a k -active sample. Firing a residual costs λ in L_0 and reduces reconstruction error by α^2 (one orthogonal component αr_i). Hence the SAE fires the residual iff $\lambda < \alpha^2$, i.e., $\lambda_{\text{c}}^{\text{Mat}} = \alpha^2$. Comparing to the vanilla bound, $\Delta \lambda_{\text{c}} = \alpha^2 - \alpha^2 k^* / (k^* - 1) = -\alpha^2 / (k^* - 1) < 0$. $\square$

D Heuristic for Proposition 4 (Archetypal shift)

Archetypal SAEs constrain each decoder atom d_j to lie in $\text{conv}(X)$, the convex hull of the training set. Under the HBG generator, any unit-norm direction reachable as a convex combination of training samples has the form $\sum_{x_t \in X} \gamma_t x_t$ with $\sum \gamma_t = 1, \gamma_t \geq 0$.

Assume the closest reachable approximation to a pure child atom $u_i = v + \alpha r_i$ under this constraint mixes other children’s residuals, picking up an expected $\alpha/\sqrt{K}$ contamination per residual direction. The corresponding reconstruction error on a k -active sample with the Archetypal version of “pure” child coding is $\alpha^2 k \cdot (k/K)$ rather than 0. Comparing costs:

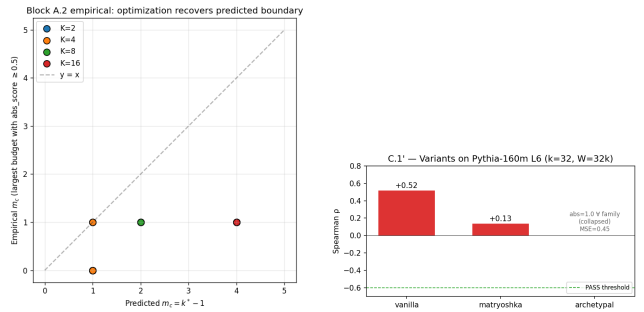
$$\mathcal{J}_{\text{child}}^{\text{Arc}} = \alpha^2 k(k/K) + \lambda k, \quad (16)$$

$$\mathcal{J}_{\text{parent}}^{\text{Arc}} = \alpha^2 k + \lambda, \quad (17)$$

$$\mathcal{J}_{\text{parent}} \leq \mathcal{J}_{\text{child}} \iff \lambda(k-1) \geq \alpha^2 k(1 - k/K), \quad (18)$$

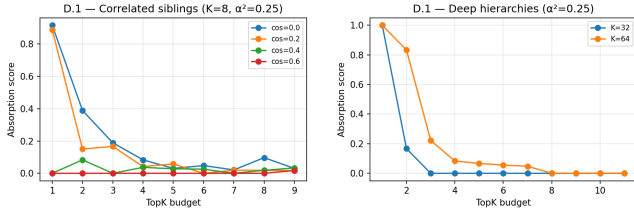
giving $\lambda_{\text{c}}^{\text{Arc}} = \alpha^2 k(1 - k/K)/(k-1)$. The shift vanishes as $K \rightarrow \infty$ (the Archetypal contamination becomes negligible relative to the $\alpha^2 k$ parent-absorption error) but is strongly downward at small K ; in the limit $k \rightarrow K$ the heuristic threshold goes to zero, consistent with the empirically observed Archetypal collapse on Pythia activations (§4).

E Synthetic verification: trained-SAE plot and variant signatures

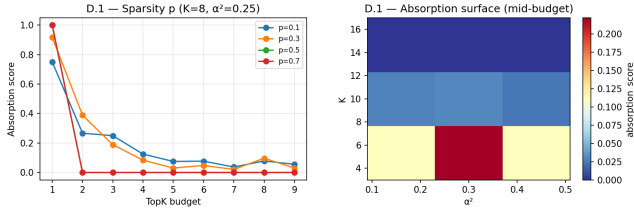


(a) Trained TopK SAE phase match (12/16 cells within ± 1 (also referenced in §4). Ma-budget; $K=16$ deviates by 3 tryoshka attenuates the wrong-direction signal; Archetypal collapses to $\text{abs_mag} = 1$.

Figure 5: Synthetic-side and Pythia-side variant evidence supporting Propositions 2–4.



(a) Correlated siblings, $c \in \{0, 0.2, 0.4, 0.6\}$ ($K=8$, $\alpha^2 = 0.25$). (b) Deep hierarchies, $K \in \{32, 64\}$ (Vanilla TopK).



(c) Sparsity sweep, $p \in \{0.1, 0.3, 0.5, 0.7\}$ ($K=8$, $\alpha^2 = 0.25$). (d) $K \times \alpha^2$ surface (mid-budget).

Figure 6: 507-row synthetic stress sweep: the bound predicts the absorption regions in all four corner regimes.

F D.1: synthetic edge-case stress sweep

G Structural-recovery detail

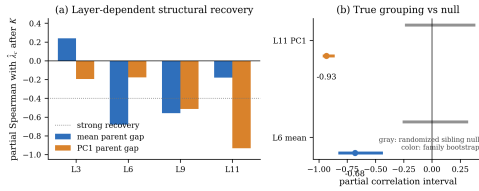
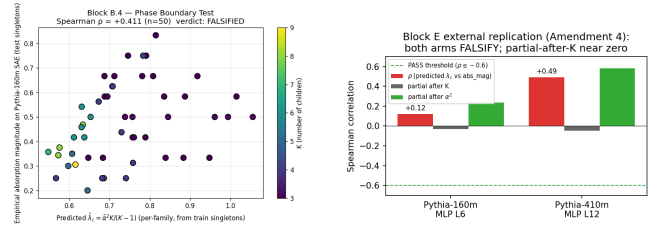


Figure 7: **Dictionary-structural evaluation recovers the hierarchy signal that the firing proxy missed.** (a) Parent-child decoder gap correlations across layers after controlling for family size K . Middle layers recover the family-mean parent direction; deep layer L11 recovers the PC1 parent direction. (b) The two main structural claims survive bootstrap intervals and are outside randomized-sibling correlation nulls.

H Pre-registered proxy and replication plots

I C.4: hedging decomposition

For each of the 50 accepted families we extract the SAE’s modal top-1 latent on the test-template singletons, read off its encoder and decoder rows ($W_{\text{enc}}[i_{\text{modal}}]$,



(a) Pre-registered Pythia-160m proxy failure. Each WordNet family; Spearman $\rho = +0.41$ out against a locked acceptance threshold of $\rho \leq -0.6$. (b) External replication from Amendment 4. Both arms fail the proxy criterion; partialling out K collapses ρ to near zero.

Figure 8: Detailed plots for the real-LLM singleton top-1 firing proxy and its external replication.

$W_{\text{dec}}[i_{\text{modal}}]$), and compute $\cos(W_{\text{enc}}, W_{\text{dec}})$ along with $\text{hedge} = 1 - |\cos|$.

- Encoder/decoder cosine: mean +0.94, median +0.94, min +0.59, max +0.99.
- Hedge score: median 0.014, max 0.41.
- Spearman ρ between abs_mag and hedge: -0.47 (higher hedging $\rightarrow$ lower abs_mag).
- Spearman ρ between λ_c and abs_mag after partialling out hedge: +0.43 (unchanged from baseline).
- Same partial after $\cos(\text{enc}, \text{dec})$: +0.43.
- Same partial after K : +0.14 (K alone explains $\sim 2/3$ of the signal).

The Pythia-160m modal latents are therefore structurally clean, with strong encoder–decoder alignment, and the spurious positive ρ at L6 is not driven by feature hedging.

J C.3: $\hat{v}$ estimator robustness

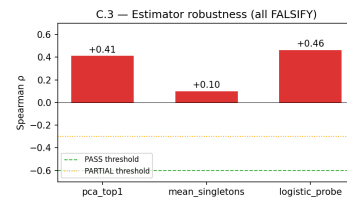


Figure 9: Three $\hat{v}$ estimators on the same 50 families; ρ remains positive for all three (spread 0.37).

K Pre-registration document (verbatim) and amendments

The locked pre-registration with all four dated amendments is reproduced verbatim in the supplementary file `PREREG_BLOCK_B.md`. Section numbering matches the preregistered-proxy supplement; the git revision at which the document was first frozen is recorded inside it.

L Three absorption notions

Notion	Operational meaning	Predicted by Theorem 1?
Structural	Does the trained dictionary contain a parent-aligned atom?	Yes (pure-strategy bound)
Per-sample firing	Does the SAE fire a child atom on a k -active sample?	No (mixed-strategy escape)
Subset skip-ping	Does a child atom fire on <i>all</i> concept members, or skip subsets?	No (Chanin notion; L_1 hedging dynamics)

Chanin et al. [2024]’s “A is for Absorption” metric targets the gerrymandered firing notion; Bussmann et al. [2025]’s reported “reduced absorption” is also a gerrymandered-firing measurement. §4’s consensus magnitude metric is per-sample firing. Only Theorem 1 targets the structural notion. Claims across notions must therefore be read against the notion they instantiate: the report that Matryoshka “reduces absorption” concerns gerrymandered firing, whereas Proposition 3’s $\lambda_c^{\text{Mat}} = \alpha^2$ concerns the structural threshold; the two do not conflict.

M Pre-Registered Instance-Level Proxy and Negative Result

Protocol. The full pre-registration is reproduced verbatim in `PREREG_BLOCK_B.md`; we summarize the locked choices here.

Hierarchy set. We select 80 noun synsets from WordNet [Miller, 1995] reachable from eight pre-registered ancestors at depth ≥ 4 , with $K \in [4, 10]$ single-token children in the Pythia tokenizer. Mechanical rejection criteria (singleton count, $\hat{\alpha}^2 < 0.001$, duplicate $\hat{v}$) leave 50 accepted families.

Model and SAE. Amendment 1 of the pre-registration switched the model from Gemma-2-2B (which requires manual license acceptance) to EleutherAI/pythia-160m; the SAE is the publicly released EleutherAI/sae-pythia-160m-32k (TopK $k = 32$, 32768 latents), hooked at the residual stream after layer 6.

Estimators (singleton-only, locked in B.2). K_i is the WordNet child count. $\hat{v}_i$ is the top right singular vector

of the centered train-template singleton matrix; $\hat{\alpha}_i^2$ is the mean residual variance after projecting out $\hat{v}_i$ divided by $d - 1$. The proxy threshold is

$$\hat{\lambda}_c^{(i)} = \frac{\hat{\alpha}_i^2 K_i}{K_i - 1}. \quad (19)$$

The theorem is stated for a sample with k active children; substituting family size K_i for the active count is therefore a modeling choice of the proxy, not a direct instantiation of the theorem.

Absorption metric (Amendment 3, consensus). For each family we feed all test-template singletons through the SAE and record the top-1 latent per sample. The absorption magnitude is the fraction of singletons whose top-1 latent equals the family’s modal top-1 latent.

Acceptance / proxy verdict. The locked verdict uses Spearman correlation between $\hat{\lambda}_c^{(i)}$ and `abs_magi` over the 50 accepted families:

$$\begin{aligned} \rho \leq -0.6 & \quad \text{PASS,} \\ \rho > -0.3 & \quad \text{FALSIFIED,} \\ -0.6 < \rho \leq -0.3 & \quad \text{PARTIAL.} \end{aligned} \quad (20)$$

The criterion is operational: it scores top-1 consensus firing rather than measuring dictionary composition directly.

Result: the proxy fails. On the locked family set ($n = 50$), the observed Spearman correlation is

$$\rho = +0.41 \quad (p = 0.003, \text{CI}_{95\%} = [+0.13, +0.62]). \quad (21)$$

The one-sided permutation p -value for the pre-registered direction (ρ at least as negative as observed, i.e., $\leq +0.41$) is 0.996 under 5,000 label permutations: only 0.4% of permutations reach a correlation this negative. The locked proxy is therefore falsified in the wrong direction. The detailed scatter appears in Appendix H; the K -driven baseline that pulls ρ positive is analyzed below.

Robustness summary. The locked verdict is stable across three robustness axes evaluated on the same family set: (i) three $\hat{v}$ estimators (PCA top-1, mean-of-singletons, 1-vs-rest logistic probe); (ii) the four cross-layer measurements (§4); (iii) an external pre-committed replication on Pythia-410m (Amendment 4). No variant reaches the $\rho \leq -0.6$ acceptance threshold: the estimators give $+0.41, +0.10, +0.46$; residual layers L3/L6/L9/L11 give $+0.34, +0.41, +0.42, -0.35$; and the pre-committed Pythia-160m/410m MLP replication gives $+0.12, +0.49$.

Diagnosis at the primary cell. The Spearman $\rho = +0.41$ decomposes via partial correlations:

$$\rho(\text{abs_mag}, K) = -0.43, \quad (22)$$

$$\rho(\text{abs_mag}, \alpha^2) = +0.10. \quad (23)$$

After subtracting the $1/K$ chance baseline from `abs_mag`, all correlations collapse to near zero. The proxy threshold

$$\lambda_c = \frac{\alpha^2 K}{K - 1} \quad (24)$$

substitutes family size for the theorem’s active count and couples K and α^2 ; in our data K dominates and pulls λ_c into the wrong-sign correlation, while α^2 itself shows no detectable signal.

Metric failure or theory failure? Two readings of the negative result are possible. The consensus metric may be contaminated by its $1/K$ chance baseline—every family of size K gives any single latent a $1/K$ probability of winning by chance—in which case the proxy probes the metric rather than the theorem; or the theory itself may fail to transfer. The diagnostic sweep in §4 separates the two readings only partially: width-invariance, which carries no K dependence, is also violated, pointing toward genuine non-transfer, whereas depth-dependence and the 410m right-direction α^2 baseline,

$$\rho_{\alpha^2} = -0.32, \quad (25)$$

indicate that part of the structure assumed by the theory survives at larger scale. We therefore characterize the finding as a *metric-mediated non-transfer*: transfer fails under the pre-registered consensus proxy and remains open under a K -baseline-corrected metric.

External replication preserves the verdict. Amendment 4 (locked 2026-05-15, before any 410m measurement) added an external replication with two arms at a matched MLP hook: Pythia-160m `layers.6.mlp` and Pythia-410m `layers.12.mlp`. Both arms fail the proxy criterion:

$$\rho_{160m} = +0.12, \quad (26)$$

$$\rho_{410m} = +0.49. \quad (27)$$

In both arms, partialling out K changes ρ by at most 0.05 and eliminates the spurious signal. The K -baseline contamination of the consensus metric is therefore the dominant confound, and it transfers across model size and hook; the replication plot appears in Appendix H.

Protocol integrity. Every verdict reported above uses the locked criteria. Amendment 4 was pre-committed to publication regardless of outcome, and no undeclared re-runs on additional models, hooks, layers, metrics, or estimators were performed in search of a passing correlation.